

Details of Feature Computation for Clustering

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February 16, 2026

1 Introduction

This document details the *feature engineering* process used to cluster drivers (license plates) based on their behavior on the highway. The goal is to transform raw vehicle flow records into a single feature vector per license plate.

2 Data Preprocessing

Before computing the variables, the flow data (F) go through a cleaning stage:

1. **License Plate Normalization:** Non-alphanumeric characters are removed and the plate is standardized to uppercase. Invalid plates (e.g., "NULL", "NAN") or plates with length outside the range $[5, 6]$ are discarded.
2. **Lane Filtering:** Only lanes 1, 2, and 3 are considered.
3. **Duplicate Removal:** Duplicate records are removed considering the tuple $(Plate, Gantry, Lane, Timestamp)$.
4. **Outlier Handling (Speed):** *Winsorization* is applied to speed for each $(Gantry, Lane)$ group, capping extreme values at the 1% and 99% percentiles.

3 Variable Computation (Event Level)

For each event i (a vehicle passing through a gantry), intermediate metrics are computed. Let v_i be the speed of event i at time t_i .

3.1 Headway (h_i)

Headway is computed as the time difference with the preceding vehicle ($i - 1$) in the **same lane and gantry**. The data must be temporally ordered.

$$h_i = t_i - t_{i-1} \quad (1)$$

If $h_i > 60$ seconds, it is assumed there is no close interaction and the value is discarded (NaN) for conflict computation purposes, although it may be included in averages if desired.

3.2 Indexed Relative Speed ($v_{rel,i}$)

To normalize behavior according to traffic conditions, the vehicle speed is compared with the average flow speed at that gantry over a 5-minute interval, denoted $\bar{V}_{5min}$.

$$v_{rel,i} = \frac{v_i}{\bar{V}_{5min}} \quad (2)$$

- $v_{rel,i} > 1$: Driver is faster than the average flow.
- $v_{rel,i} < 1$: Driver is slower than the average flow.

3.3 Conflict and TTC (Time To Collision)

Collision risk with the preceding vehicle is evaluated. TTC_i is computed only if vehicle i is traveling faster than the previous vehicle ($v_i > v_{i-1}$) and the headway is valid.

$$TTC_i = \frac{h_i \cdot v_i}{v_i - v_{i-1}} \quad (3)$$

A **conflict** event (c_i) is defined if TTC_i is below a gantry-specific threshold (TTC_{max}), defined empirically.

$$c_i = \begin{cases} 1 & \text{if } TTC_i < TTC_{max} \\ 0 & \text{if } TTC_i \geq TTC_{max} \text{ or } v_i \leq v_{i-1} \end{cases} \quad (4)$$

4 Aggregation by License Plate

The goal is to obtain a single vector per driver P . The aggregation process can be performed in two ways: Global or Monthly-Weighted.

Let N_P be the total number of recorded passes for license plate P .

4.1 Aggregated Variables

4.1.1 Average Speed ($\bar{v}_P$)

$$\bar{v}_P = \frac{1}{N_P} \sum_{i \in P} v_i \quad (5)$$

4.1.2 Average Relative Speed ($\bar{v}_{rel,P}$)

$$\bar{v}_{rel,P} = \frac{1}{N_{rel}} \sum_{i \in P} v_{rel,i} \quad (6)$$

Where N_{rel} is the number of events with valid relative speed.

4.1.3 Average Headway ($\bar{h}_P$)

$$\bar{h}_P = \frac{1}{N_h} \sum_{i \in P, h_i \leq 60} h_i \quad (7)$$

4.1.4 Conflict Rate (CR_P)

Proportion of events with valid headway that resulted in a conflict.

$$CR_P = \frac{\sum_{i \in P} c_i}{N_h} \quad (8)$$

4.1.5 Lane Usage ($Lane_k$)

Proportion of usage of lane $k \in \{1, 2, 3\}$.

$$Lane_k = \frac{\text{Count of passes in lane } k}{N_P} \quad (9)$$

4.1.6 Lane Change Rate (LCR_P)

Estimated by counting how many times the vehicle changed lanes between consecutive gantries. Let L_j be the lane in event j (time-ordered).

$$Changes = \sum_{j=2}^{N_P} \mathbb{I}(L_j \neq L_{j-1}) \quad (10)$$

$$LCR_P = \frac{Changes}{N_P - 1} \quad (11)$$

Note: This metric is approximate, as it assumes continuity between recorded gantries.

4.2 Consolidation Strategies

4.2.1 1. Global Aggregation (Default)

The formulas above are applied over **all** historical records for license plate P as a single set.

4.2.2 2. Monthly Weighting

Monthly averages $X_{P,m}$ are computed first for each month m , and then weighted by the number of trips in that month ($N_{P,m}$).

Let X be any variable (e.g., Speed, Conflict Rate). The final weighted value is:

$$X_{P,final} = \frac{\sum_m (X_{P,m} \cdot N_{P,m})}{\sum_m N_{P,m}} \quad (12)$$

This ensures that months with higher activity have greater influence, while still allowing intermediate monthly metrics if temporal analysis is required.

5 Definition of Frequent Drivers

For clustering model training, users are separated into:

- **Frequent:** Users with enough history to characterize their behavior. Typical criteria:
 - $N_P \geq 20$ total passes.
 - Active days ≥ 5 .
- **Infrequent:** Users with few data points. They are not used for training, but can be assigned to a cluster later using the trained model (or remain as "Unknown").

6 Python Implementation

The computation logic described above is implemented mainly in the module `src/clustering.py`. Below, we detail the execution flow of the key functions for auditing purposes.

6.1 Data Cleaning: `clean_flujos_for_clustering`

This function takes the raw flow DataFrame and performs the initial preparation:

1. **Column Validation:** Ensures that essential columns exist (timestamp, speed, gantry, lane, license plate).

2. **License Plate Cleaning:** Plates are normalized (alphanumeric, uppercase) and filtered if their length is invalid (outside the range [5, 6]).
3. **Basic Filtering:** Rows with null values in critical fields are removed and only lanes 1, 2, and 3 are kept.
4. **Deduplication:** Exact duplicates of the tuple (Plate, Gantry, Lane, Timestamp) are removed. If duplicates exist with different speeds, a deterministic priority is applied (merge sort ordering preserving the first).
5. **Winsorization:** Extreme speed values are capped by group (Gantry, Lane) using the defined percentiles (default 1% and 99%).

6.2 Feature Computation: Clusterization

This is the main function that orchestrates feature computation.

6.2.1 1. Simple Metrics and Pre-computation

Metrics that do not require complex temporal ordering are computed:

- **Total Passes** (`total_passes`): Simple count of records per license plate.
- **Average Speed** (`avg_speed_kmh`): Arithmetic mean of speed.
- **Lane Usage:** A pivot table is built to count passes per lane and divide by the total.

6.2.2 2. Relative Speed

- The average flow speed is computed in 5-minute windows per gantry (`interval_speed_mean`).
- This average is joined to each event to compute `relative_speed` = $v_i / \bar{V}_{5min}$.

6.2.3 3. Interaction Metrics (Headway, Conflict)

This stage is critical and sensitive to data ordering:

1. **Sorting:** Data are strictly sorted by Gantry, Lane, and Timestamp.
2. **Vectorized Operations (Shift):** Shifted columns (`shift(1)`) are created to compare the current event with the immediately preceding vehicle in the same lane/gantry.
3. **Headway Computation:** Time difference between the current and previous event. Headways greater than `max_headway_s` (60s) are discarded (NaN).
4. **Conflict Computation:**
 - Compute $TTC = \frac{Headway \cdot v_{current}}{v_{current} - v_{previous}}$.
 - Valid only if $v_{current} > v_{previous}$.
 - Mark conflict if $TTC < TTC_{max}$ (where TTC_{max} depends on the gantry).
5. **Aggregation:** Valid events are summed and counted by license plate to obtain averages.

6.2.4 4. Lane Changes (Lane Changes)

- Data are sorted by License Plate and Timestamp.
- The current lane is compared with the previous one. If they differ (and it is the same plate), it is counted as a change.
- The rate is normalized by $(N_P - 1)$.

6.2.5 5. Weighting (If applicable)

If `monthly_weighting` is enabled, metrics are computed first by grouping by Plate-Month, and then a weighted average is taken using the number of trips in the month to obtain the final value for the plate.